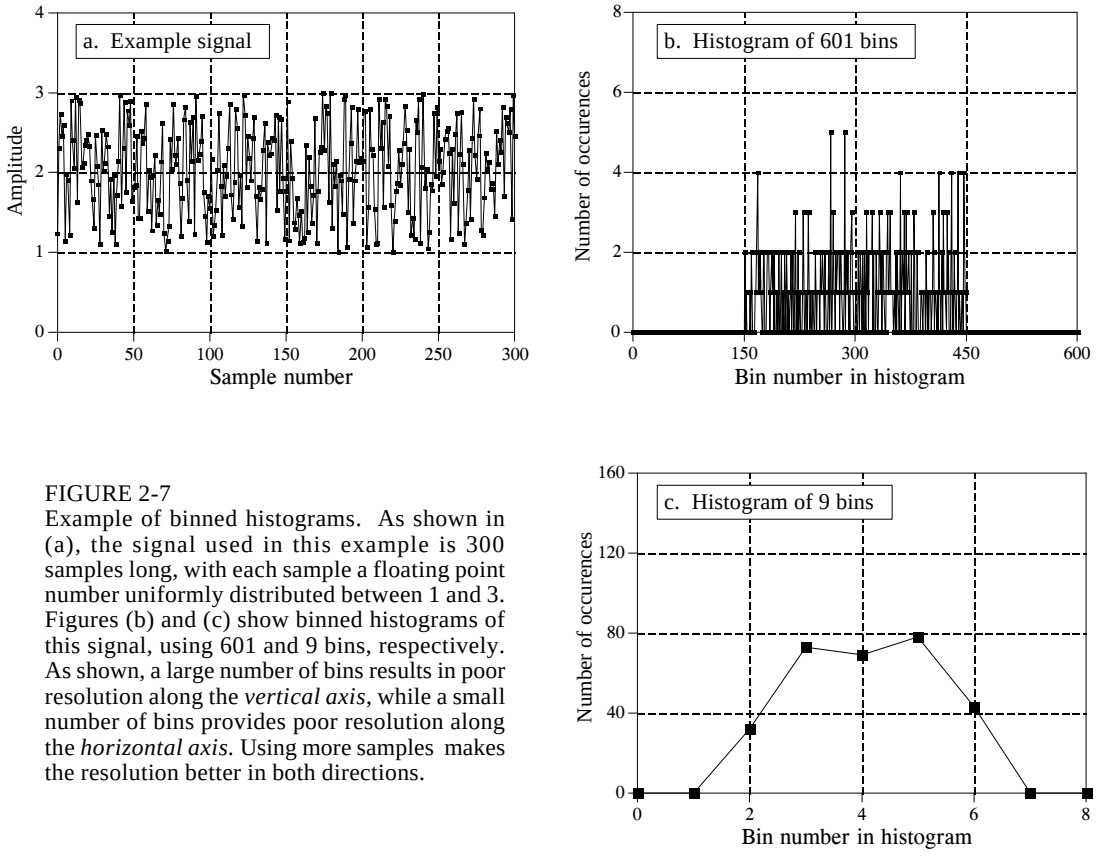

FIGURE 2-7
Example of binned histograms. As shown in (a), the signal used in this example is 300 samples long, with each sample a floating point number uniformly distributed between 1 and 3. Figures (b) and (c) show binned histograms of this signal, using 601 and 9 bins, respectively. As shown, a large number of bins results in poor resolution along the *vertical axis*, while a small number of bins provides poor resolution along the *horizontal axis*. Using more samples makes the resolution better in both directions.

How many bins should be used? This is a compromise between two problems. As shown in Fig. 2-7, too many bins makes it difficult to estimate the *amplitude* of the underlying pmf. This is because only a few samples fall into each bin, making the statistical noise very high. At the other extreme, too few of bins makes it difficult to estimate the underlying pmf in the *horizontal* direction. In other words, the number of bins controls a tradeoff between resolution in along the y-axis, and resolution along the x-axis.

The Normal Distribution

Signals formed from random processes usually have a bell shaped pdf. This is called a **normal distribution**, a **Gauss distribution**, or a **Gaussian**, after the great German mathematician, Karl Friedrich Gauss (1777-1855). The reason why this curve occurs so frequently in nature will be discussed shortly in conjunction with *digital noise generation*. The basic shape of the curve is generated from a *negative squared exponent*:

$$y(x) = e^{-x^2}$$